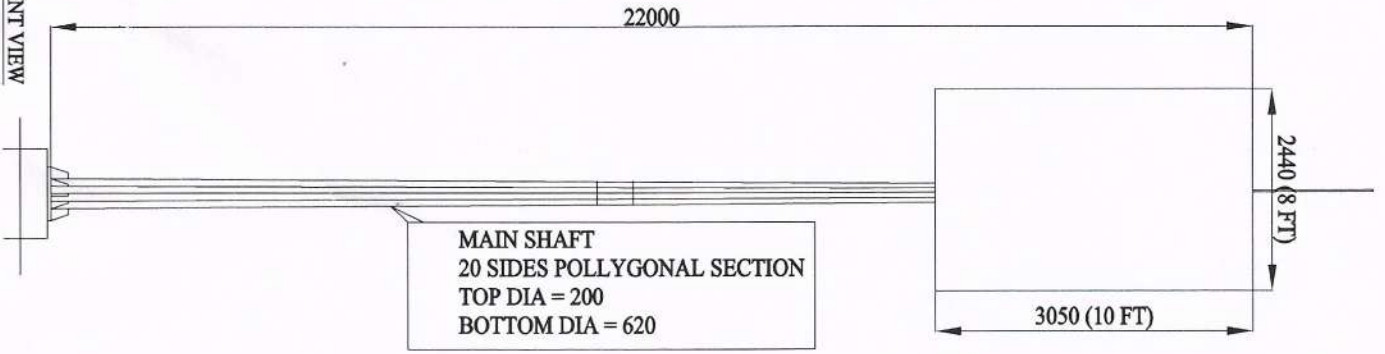


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KOLKATA - 700006**

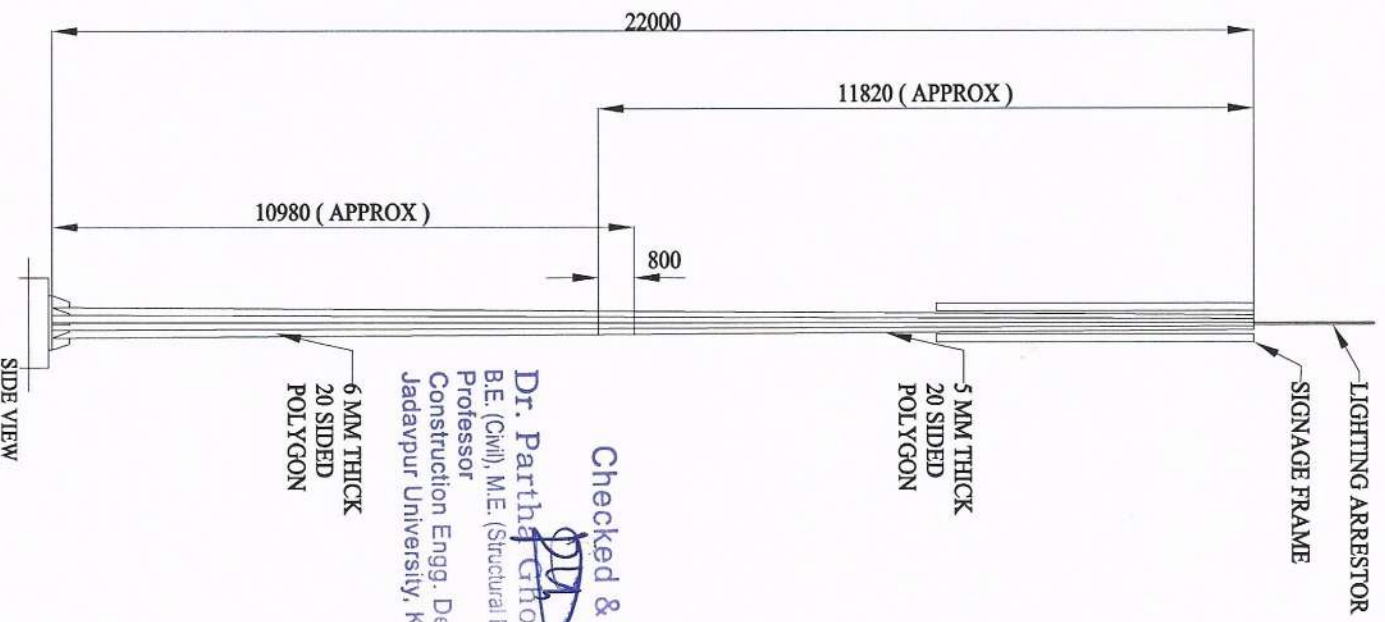
CLIENT: BHARAT PETROLEUM CORPORATION LIMITED
PROJECT TITLE:
STRUCTURAL DESIGN CALCULATION FOR 22M
SIGNAGE MAST WITH INLAND LOGO FRAME
(WIND SPEED : 50m/s; SBC : 4T/SQM)

FRONT VIEW



MAIN SHAFT
20 SIDES POLLYGONAL SECTION
TOP DIA = 200
BOTTOM DIA = 620

SIDE VIEW



LIGHTING ARRESTOR
SIGNAGE FRAME

5 MM THICK
20 SIDED
POLYGON

6 MM THICK
20 SIDED
POLYGON

Checked & Vetted

Dr. Partha Ghosh
B.E. (Cvill), M.E. (Structural Engg.), Ph.D. (Engn ISO-1461:1999 AND IS 11759-1985.
Professor
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Jadavpur University, Kolkata-700

NOTE :- 1. ALL DIMENSION ARE IN MM.

2. WIND SPEED = 50 M/ SEC.

3. DIMENSION TOLERANCE BSEN 40

4. MATERIAL TOLERANCE IS 1852-1985

5. MATERIAL PROTECTION = HOT DIP GALVANIZED

6. GALVANIZING AND FINISHING STANDERD

7. SHAFT MATERIAL TYPE BSEN 10025/EQUIVALENT.

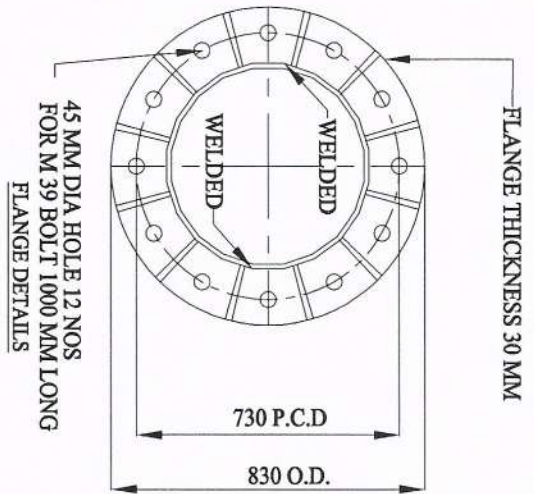
8. FLANGE MATERIAL IS 2062.

9. SECTION LENGTH MAY CHANGE DUE TO

10. AVAILABILITY OF MAT.

10. NUMBER OF LOGITUDINAL WELD JOINT WILL BE 2 OR

AS PER AVAILABILITY OF WIDTH OF THE MATERIAL.



FLANGE THICKNESS 30 MM

45 MM DIA HOLE 12 NOS
FOR M 39 BOLT 1000 MM LONG
FLANGE DETAILS

B.P. PROJECTS PVT LTD.

167 A, VIVEKANANDA ROAD • KOL-700006

BPP

HIGH
MAST

PROJECT :- BPCL

TITLE :- 22 MTR. HIGH MAST SIGNAGE

SCALE: N.T.S

REV. NO.

SHEET NO.

DESIGN	NAME	DATE

DRAWING	G. P.	1004/25
CHECKED	G. K.	1004/25

APPROVED	BPP	

DRN NO. :-	BP-22 M-SIGN-620-200
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BPP HIGH MAST

TYPE OF STRUCTURE :

The superstructure of the Logo Mast is a slender column with hollow 20 sided polygonal section, fixed at its base and free at the top, and the diameter is reducing from its base to top. The mast is made of high strength steel metal in two parts, and joined together in SLIP Jointing system. The bottom part is welded with a base plate along with stiffeners at its lower end.

The foundation is of Reinforced Cement Concrete having a square base with a square pedestal at centre.

The superstructure is fixed with the foundation by means of foundation bolts in required numbers.

SCOPE OF WORK :

The entire structure is analysed and designed in two parts, one is the superstructure and the other is the foundation. Our scope of work is to analysis and design the Superstructure in PART- A and the R.C.C. foundation in PART- B.

PART - A

STRUCTURAL DESIGN CALCULATIONS FOR THE STEEL MAST

DESIGN BASIS :

As the dead load is very small compared to its geometry, the design seismic Lateral forces is small, and not considered for this design. But, due to larger height in comparison to its sectional dimension, effects of the lateral forces exerted by wind on the mast will be the governing lateral load for design.

The following code of practices are considered for design calculations.

- (1) IS:800-2007
- (2) IS:806-1968
- (3) IS:875 (Part 3)-2015
- (4) IS:1367 (Part 3)-2002
- (5) IS:9595-1996
- (6) IS:10178-1995
- (7) BS:5649 (Part 7)-1985
- (8) IS:1904-1986
- (9) IS:11233-1985
- (10) IS:6533 (Part 2)-1989

**BPP
HIGH MAST**

DESIGN PARAMETERS :

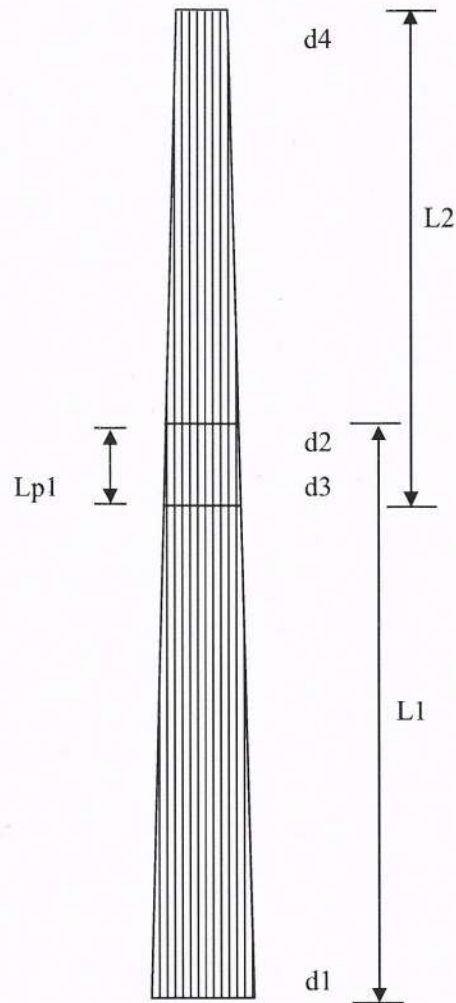
1. Total Height of Logo Mast	= 22 m	
2. Total No. of Sections	= 2	
3. Dimension of bottom Dia.(Outer)	= 620 mm	
4. Dimension of top Dia.(Outer)	= 200 mm	
5. Lap Length (Ref: Page 5)	= 800 mm	
6. Joining of section	: Telescopic, continuous & Tapering	
7. Section Type	: Regular Polygon of 20 sides	
8. Thickness of lower section	= 6 mm	
9. Thickness of top section	= 5 mm	
10. No.of Logo Signage fixed at top	= 2 nos.	
11. Wind Blokage Area of conical part of mast (Ref: Page 7)	= 7.82 sqm	
12. Wind Blokage Area of Logo Signage & fittings etc. (8ft x 10ft i.e. 2.44 x 3.05 m2) (Ref: Page 6)	= 7.4 sqm	
13. Self weight of Mast (Shell portion only) (Ref: Page 5)	= 1283.23 Kg	(approx.)
14. Weight of base plate & stiffeners (Ref: Page 5)	= 63 Kg	(approx.)
15. Extra Component (Fixing Clit, Channels , Top Plate Cover, Lig	= 35 Kg	(approx.)
16. Weight of Logo fixing frame and its Nuts & bolts (Ref: Page 5)	= 257.5 Kg	(approx.)
17. Weight of ACM Board sheet (Ref: Page 5)	= 80 Kg	(approx.)
18. Material of construction	BS EN 10027 S355	
19. Grade of foundation concrete	= M20	
20. Grade of foundation steel	= Fe500	
21. Load Bearing Capacity of soil	= 4.0 T/m2	
22. Base plate size	= 830 mm dia x 30 mm thk	
23. Foundation Bolts	= 12 nos. = 39 mm dia x 1000 mm long at 730 mm PCD	
24. Wind speed	= 50 m/s i.e. 180 km/hr	
25. Terrain category	= 2	
26. Mean design life of structure	= 25 years	

Checked & Vetted


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Professor
Construction Engg. Department
Jadavpur University, Kolkata-700 106

**BPP
HIGH MAST**

GEOMETRY DETAILS :



Total Length, L	= 22 m
No. of segments	= 2
L1	= 10.980 m
L2	= 11.820 m
t1 (Thickness of 'L1' segment)	= 0.006 m
t2 (Thickness of 'L2' segment)	= 0.005 m
d1	= 0.620 m
d4	= 0.200 m

BPP HIGH MAST

$$\text{Change of dia. per unit length, } c = 0.019$$

$$L_{p1} = 0.800 \text{ m}$$

$$d_2 = d_1 - c \cdot L_1 = 0.410 \text{ m}$$

$$d_3 = d_2 + L_{p1} \cdot c + 2t_2 = 0.436 \text{ m}$$

CALCULATION FOR DEAD LOAD :

a) Self Weight of Mast

$$= 7850 \cdot \pi \cdot 0.5 \cdot [(L_1 \cdot t_1 \cdot (d_1 + d_2 - 2t_1)) + (L_2 \cdot t_2 \cdot (d_3 + d_4 - 2t_2))]$$

$$= 1283 \text{ Kg}$$

b) Weight of ACM Board sheet (2.44 x 3.05 m2) = 80 Kg (Approx.)

c) Weight of Logo fixing frame and its Nuts and bolts = 257.5 Kg (Approx.)

d) Extra Component (Fixing Clit, Channels , Top Plate Cover, Ligl= 35 Kg (Approx.)

e) Weight of base plate & stiffeners = 63 Kg (Approx.)

Total applied Dead Load at Mast top = 372.2 Kg

Total Dead Load at Mast Base = 1718.11 Kg

Total Mass, m = 17181.1 N

BPP HIGH MAST

CALCULATION FOR WIND LOAD :

Vb=Basic Wind Speed, (Ref: Wind Map of IS: 875 (Part 3)- 2015)	= 50 m/s
Vz=Design Wind Speed = $k_1 * k_2 * k_3 * k_4 * V_b$ (Ref: Clause 6.3 of IS: 875 (Part 3)- 2015)	
k1=Probability Factor (Ref: Table-1 of clause 6.3.1 of IS: 875 (Part 3)- 2015)	= 0.90
k2=Terrain and Height Factor (Ref: Table-2 of clause 6.3.2.2 of IS: 875 (Part 3)- 2015)	
Terrain category	= 2
k3=Topography Factor (Ref: Cl.6.3.3 of IS: 875 (Part 3)- 2015)	= 1.0
k4=Importance factor for cyclonic region for all other structures (Ref: Cl.6.3.4 of IS: 875 (Part 3)- 2015)	= 1.0
Pz=Design Wind Pressure= $0.6 * V_z^2$ N/m ² (Ref: Cl.7.2 of IS: 875 (Part 3)- 2015)	
Pd=Design Wind Pressure= $K_d \times K_a \times K_c \times P_z$ N/m ² =	
Kd = Wind directionality factor =	= 1.0
Ka = Area averaging factor =	= 1.0
Kc = Combination factor =	= 1.0
Wind Force, Fx= $A * K * C_f * P_z$ where, A=Wind Blockage Area	
Cf=Shape Factor (Ref: Cl.7.4.2 of IS: 875 (Part 3)- 2015)	
ϵ/D =Relative Surface Roughness (Ref: Note in Appendix-D of IS: 875 (Part 3)- 2015)	= 0.001
ν =Kinematic viscosity of air (Ref: Note in Appendix-D of IS: 875 (Part 3)- 2015)	= $1.46 * 10^{-5}$ m ² /s
Change of dia. per unit length, c= (Dtop - Dbot)/L	= 0.019
L/D	= 54.881
Dvd	= 18.921
	> 6.0 m ² /s
K=Reduction Factor (Ref: Table-26 & FIG.5 of IS: 875 (Part 3)- 2015)	= 0.991

Calculation of Wind Blockage Area at Top :

No.of Logo Signage fixed at top	= 2 nos.
No.of Logo Signage facing wind pressure (in worst condition)	= 1 nos.
Size of each Logo Signage (2.44 x 3.05 m ²)	= 7.44 m ²
Total wind Blockage Area at Top	= 7.44 m ²
The C.G. of Logo Signage fittings are placed at a distance from top	= 0.8 m

**BPP
HIGH MAST**

CALCULATION FOR NODAL FORCES AND MOMENTS DUE TO WIND:

Node Ht. (m)	Top Dia (m)	Bot Dia (m)	Avg. Dia. (m)	Windage Area (m ²)	K ₂	V _z (m/s)	Re=(D*V _z)/ ν (in 10 ⁻⁵)	Cf/(1+2e/D) (Fig.5)	K*Cf	P _z (T/m ²)	F, Wind Force at node (T)	M _z , Mom about base (T-m)	M _z , Mom at node (T-m)
22		0.200	0.200	7.442	1.2	54.00	7.4	0.581	0.581	0.1750	0.757	16.047	
21	0.200	0.219	0.210	0.210	1.2	54.00	7.8	0.606	0.606	0.1750	0.022	0.467	0.151
20	0.219	0.238	0.229	0.229	1.2	54.00	8.5	0.650	0.648	0.1750	0.026	0.519	0.931
19	0.238	0.257	0.248	0.248	1.2	54.00	9.2	0.694	0.691	0.1750	0.030	0.569	1.736
18	0.257	0.276	0.267	0.267	1.062	47.79	8.7	0.669	0.664	0.1370	0.024	0.437	2.571
17	0.276	0.295	0.286	0.286	1.058	47.61	9.3	0.706	0.701	0.1360	0.027	0.463	3.430
16	0.295	0.315	0.305	0.305	1.054	47.43	9.9	0.744	0.737	0.1350	0.030	0.485	4.317
15	0.315	0.334	0.324	0.324	1.05	47.25	10.5	0.762	0.751	0.1340	0.033	0.489	5.233
14	0.334	0.353	0.343	0.343	1.04	46.80	11.0	0.780	0.764	0.1314	0.034	0.483	6.183
13	0.353	0.372	0.362	0.362	1.03	46.35	11.5	0.795	0.763	0.1289	0.036	0.463	7.167
12	0.372	0.391	0.381	0.381	1.02	45.90	12.0	0.807	0.762	0.1264	0.037	0.441	8.187
11	0.391	0.410	0.400	0.400	1.01	45.45	12.5	0.822	0.763	0.1239	0.038	0.417	9.243
10	0.410	0.429	0.420	0.420	1	45.00	12.9	0.837	0.763	0.1215	0.039	0.389	10.337
9	0.429	0.448	0.439	0.439	1	45.00	13.5	0.855	0.770	0.1215	0.041	0.369	11.470
8	0.448	0.467	0.458	0.458	1	45.00	14.1	0.873	0.765	0.1215	0.043	0.340	12.644
7	0.467	0.486	0.477	0.477	1	45.00	14.7	0.888	0.757	0.1215	0.044	0.307	13.861
6	0.486	0.505	0.496	0.496	1	45.00	15.3	0.901	0.754	0.1215	0.045	0.272	15.122
5	0.505	0.525	0.515	0.515	1	45.00	15.9	0.906	0.725	0.1215	0.045	0.227	16.427
4	0.525	0.544	0.534	0.534	1	45.00	16.5	0.910	0.728	0.1215	0.047	0.189	17.779
3	0.544	0.563	0.553	0.553	1	45.00	17.1	0.914	0.731	0.1215	0.049	0.147	19.177
2	0.563	0.582	0.572	0.572	1	45.00	17.6	0.919	0.735	0.1215	0.051	0.102	20.625
1	0.582	0.601	0.591	0.591	1	45.00	18.2	0.923	0.738	0.1215	0.053	0.053	22.124
0	0.601	0.620	0.610	0.610	1	45.00	18.8	0.927	0.742	0.1215	0.055	0.000	23.675

Table 26 Force Coefficients for Low Walls or Hoardings (< 15m High)
(Clause 7.4.2.2)

WIDTH TO HEIGHT RATIO, b/h		DRAG COEFFICIENT C _f
WALL ABOVE GROUND	WALL ON GROUND	
FROM 0.5 TO 6	FROM 1 TO 12	1.2
Total =		1.607
		23.675

**BPP
HIGH MAST**

DESIGN CHECK / CALCULATION

CHECK FOR CRITICAL DEAD LOAD

Total Applied Load at Top	= 3722 N
Modulus of Elasticity of steel, E	= 200000 N/mm ²
Outer dia, D _{top}	= 200 mm
Outer dia, D _{bot}	= 620 mm
Avg. outer dia, D _{ave}	= 410 mm
Inner dia, d _{top}	= 190 mm
Inner dia, d _{bot}	= 608 mm
Avg. inner dia, d _{ave}	= 399 mm
Moment of Inertia, I _{ave} = $\pi(D_{ave}^4 - d_{ave}^4)/64$	= 142974451 mm ⁴
Total length, L	= 22000 mm
P _{crit} = $\pi^2 E I_{ave} / (4L^2)$	= 145775 N >
Applied load at top	= 3722 N
Total Dead Load on mast	= 17181.1 N
	Hence OK

Section classification :

(Ref: Cl. 3.7 of IS: 800 - 2007)

$$\varepsilon = (250/f_y)^{0.5} = (250/355)^{0.5} = 0.84$$
$$D/t = 620/6 = 103.33 > 88\varepsilon^2$$

Hence, the section is semi-compact and full cross section is available

(Ref: Cl. 3.7.2 & 3.7.4, Table-2 of IS: 800 - 2007)

CHECK FOR COMPRESSION RESISTANCE :

(Ref: Cl. 6.1 & 9.3.1.1 of IS: 800 - 2007)

Max. factored compressive force, F _{x,max}	= 25.77 KN
---	------------

The design compression resistance of the cross section,

N _d = A _g * f _y / γ _{m0} = 143.03 * 355 / 1.1 =	= 46.16 KN
	> 25.77 KN

Hence O.K.

Gross area of cross-section, A _g	= 143.0 mm ²
---	-------------------------

Yield strength of steel, f _y	= 355 N/mm ²
---	-------------------------

Partial safety factor, γ _{m0}	= 1.10
--	--------

BPP HIGH MAST

CHECK FOR DIRECT AXIAL STRESS

(Considering Average Section)

Grade of Steel = Fe 510
Yield Stress of Steel, f_y (Ref: Table-3 of IS: 5986-2002) = 355 Mpa

Allowable Stress in direct compression = σ_{ac} (Ref: Cl.11.3 of IS: 800-2007) = Minimum of

a) $0.6f_y$ = 213 Mpa

b) $\sigma_{ac} = 0.6(f_{cc} * f_y) / (f_{cc}^n + f_y^n)^{1/n}$

Where, $f_{cc} = \pi^2 * E / (\lambda)^2$

n = 1.4

$l = 2L$ = 44000 mm

r , (for hollow section) = $r_{ave} = \sqrt{I/A}$ = 143.03 mm

$\lambda = l/r$ = 307.64

f_{cc} (cal) = 20.86 Mpa

σ_{ac} (as per the above formula or Refer, Table- 11.3 of IS: 800 - 2007) = 12.35 Mpa

Hence Permissible value of " σ_{ac} " at mast base = 12.35 Mpa

Cross sectional area (calculated on the basis of average dia.) = $\pi * (D_{ave}^2 - d_{ave}^2) / 4$
= 6989.26 mm²

σ_{ac} (cal) = Total Dead Load at Mast Base / Cross Sectional Area = 2.46 Mpa <
= 12.35 Mpa
Hence OK

CHECK FOR COMBINED BENDING AND AXIAL LOAD

(At Base of the mast)

Max. Axial Load at base (DL), F_y = 17181.1 N

Max. Horizontal Load at base (WL), F_x = 16068.3 N

Max. Moment, (WL+DL), M = 236754453.5 Nmm

Sectional Area of Mast at base, $A = \pi * (D^2 - d^2) / 4$ = 11573.63 mm²

Sectional Moment of Inertia at base, $I = \pi * (D^4 - d^4) / 64$ = 545453482.71 mm⁴

At base, $y = D/2$ = 310.00 mm

Section Modulus, $z = I/y$ = 1759527.36 mm³

Direct Stress, σ_{ac} (cal) = F_y/A = 1.48 N/mm²

Bending stress, σ_{bc} (cal) = M/z = 134.56 N/mm²

h_e = effective length = L = 22 m

D = Mean Diameter at Mast Base = $(D_{bot} + d_{bot})/2$ = 0.614 m

h_e/D = 35.831

$A = [0.84 + (0.019 * h_e/D)^2]^{-1}$ for $h_e/D > 21$

And $A = 1$ for $h_e/D < 21$

Hence, A = 0.767

Mast plate thickness at base = t = 0.006 m

BPP HIGH MAST

D/t	= 102.33
B=270*t/D*(1-67*t/D) for D/t>130	
And B=1 for D/t<130	
Hence, B	= 1.000
(Ref: Annex C of IS: 6533 (Part 2)- 1989)	
Maximum permissible compressive stress in combined action of bending and direct load =	
$\sigma_{bc}=0.5f_y*A*B$	= 136.176 Mpa
For wind load (Ref: Cl.11.4 of IS: 800- 2007)	
Combined σ_{bc} (perm) = 1.33 * σ_{bc} =	= 181.114 Mpa
σ_{bt} (perm)= 1.33*0.66 f_y (Ref: Cl.11.4 of IS: 800- 2007)	= 311.619 Mpa
Combined Stress a) Max.	= 136.040 Mpa
	Compression < σ_{bc} (Perm)
b) Min.	= 133.071 Mpa
	Tension < σ_{bt} (Perm)

CHECK FOR COMBINED BENDING AND AXIAL LOAD

(At top of 1st Lap)

Max. Axial Load at base (DL), F_{y1}	= 7100.8 N
Max. Horizontal Load at base (WL), F_{x1}	= 11331.3 N
Max. Moment, (WL+DL), M_1	= 103370848.9 Nmm
Sectional Area of Mast at 1st Lap, $A_1=\pi*[(d_2+2*t_2)^2-(d_2)^2]/4$	= 6524.80 mm ²
Sectional Moment of Inertia at 1st Lap, $I_1=\pi*[(d_2+2*t_2)^4-(d_2)^4]/64$	= 140745740.48 mm ⁴
At 1st Lap, $y_1=(d_2+2*t_2)/2$	= 210.19 mm
Section Modulus, $z_1=I_1/y_1$	= 669609.08 mm ³
Direct Stress, σ_{ac} (cal)= F_{y1}/A_1	= 1.09 N/mm ²
Bending stress, σ_{bc} (cal)= M_1/z_1	= 154.37 N/mm ²

h_e = effective length = L-L ₁	= 11.020 m
D= Mean Diameter at 1st Lap= $(d_2+2*t_2+d_2)/2$	= 0.415 m
h_e/D	= 26.530
$A=[0.84+(0.019*h_e/D)^2]^{-1}$ for $h_e/D>21$	
And $A=1$ for $h_e/D<21$	
Hence, A	= 0.914

Mast plate thickness at 1st Lap= t_2	= 0.005 m
D/ t_2	= 83.08

B=270*t/D*(1-67*t/D) for D/t>130
And B=1 for D/t<130

Hence, B	= 1.000
(Ref: Annex C of IS: 6533 (Part 2)- 1989)	

Maximum permissible compressive stress in combined action of bending and direct load =	
$\sigma_{bc}=0.5f_y*A*B$	= 162.236 Mpa

For wind load (Ref: Cl.11.4 of IS: 800- 2007)

Combined σ_{bc} (perm) = 1.33 * σ_{bc} =	= 215.774 Mpa
σ_{bt} (perm)= 1.33*0.66 f_y (Ref: Cl.11.4 of IS: 800- 2007)	= 311.619 Mpa

BPP HIGH MAST

Combined Stress a) Max.

= 155.463 Mpa

Compression $< \sigma_{bc}(\text{Perm})$

b) Min.

= 153.287 Mpa

Tension $< \sigma_{bt}(\text{Perm})$

CHECK FOR COMBINED STRESS RATIO (AXIAL COMPRESSION AND BENDING)

(Ref: Cl.11.5.2 of IS: 800- 2007)

(At Base of the mast)

Allowable Stress in direct compression = σ_{ac} (Ref: Cl.11.3 of IS: 800-2007) = Minimum of

a) $0.6f_y$

= 213 Mpa

b) $\sigma_{ac} = 0.6(f_{cc} * f_y) / (f_{cc}^n + f_y^n)^{1/n}$

Where, $f_{cc} = \pi^2 * E / (\lambda)^2$

n

= 1.4

$l = 2L$

= 44000 mm

r, (for hollow section) = $r_{ave} = \sqrt{I/A}$

= 217.09 mm

$\lambda = l/r$

= 202.68

$f_{cc} \text{ (cal)}$

= 48.05 Mpa

σ_{ac} (as per the above formula or Refer, Table- 11.3 of IS: 800- 2007)

= 27.64 Mpa

Hence Permissible value of " σ_{ac} " at mast base

= 27.64 Mpa

$\sigma_{bc}(\text{perm}) = 1.33 * 0.66f_y$ (Ref: Cl.11.4 of IS: 800- 2007)

= 234.300 Mpa

$\sigma_{ac}(\text{cal}) / \sigma_{ac}(\text{perm})$

= 0.054

$N = \text{Combined Stress Ratio} = \sigma_{ac}(\text{cal}) / \sigma_{ac}(\text{perm}) + C_m * \sigma_{bc}(\text{cal}) / \{1 - \sigma_{ac}(\text{cal}) / 0.6f_{cc} * \sigma_{bc}(\text{perm})\}$

= 0.85

C_m , (as per Cl.11.5.2 of IS: 800- 2007)

= 0.73

N=

< 1.0, Hence O.K.

CHECK FOR COMBINED STRESS RATIO (AXIAL COMPRESSION AND BENDING)

(Ref: Cl.11.5.2 of IS: 800- 2007)

(At top of 1st Lap)

Allowable Stress in direct compression = σ_{ac} (Ref: Cl.11.3 of IS: 800-2007) = Minimum of

a) $0.6f_y$

= 213 Mpa

b) $\sigma_{ac} = 0.6(f_{cc} * f_y) / (f_{cc}^n + f_y^n)^{1/n}$

Where, $f_{cc} = \pi^2 * E / (\lambda)^2$

n

= 1.4

$l = 2L$

= 22040 mm

r, (for hollow section) = $r_{ave} = \sqrt{I/A}$

= 146.87 mm

$\lambda = l/r$

= 150.06

$f_{cc} \text{ (cal)}$

= 87.65 Mpa

σ_{ac} (as per the above formula or Refer, Table- 11.3 of IS: 800- 2007)

= 47.86 Mpa

Hence Permissible value of " σ_{ac} " at mast base

= 47.86 Mpa

$\sigma_{bc}(\text{perm}) = 1.33 * 0.66f_y$ (Ref: Cl.11.4 of IS: 800- 2007)

= 234.300 Mpa

$\sigma_{ac}(\text{cal}) / \sigma_{ac}(\text{perm})$

= 0.023

$N = \text{Combined Stress Ratio} = \sigma_{ac}(\text{cal}) / \sigma_{ac}(\text{perm}) + C_m * \sigma_{bc}(\text{cal}) / \{1 - \sigma_{ac}(\text{cal}) / 0.6f_{cc} * \sigma_{bc}(\text{perm})\}$

= 0.85

C_m , (as per Cl.11.5.2 of IS: 800- 2007)

= 0.65

N=

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< 1.0, Hence O.K.

CHECK FOR LAP LENGTH

(1st Lap)

Lap Length, $L_1 = (3 \cdot M_1 / \sigma_y \cdot t_1)^{0.5}$

Moment at 1st Lap=

= 103370848.9 Nmm

σ_y

= 213.000 Mpa

t_1

= 6 mm

L_1

= 493 mm

Lap Length provided, L_{p1}

= 800 mm

Hence O.K.

CHECK FOR EQUIVALENT STRESS (SHEAR AND BENDING)

(Ref: Cl.11.5.1 of IS: 800- 2007)

Max. Axial Load at base (DL), F_y

= 17181.1 N

Max. Horizontal Load at base (WL), F_x

= 16068.3 N

Max. Moment, (WL+DL), M

= 236754453.5 N

Wind force, $F(\text{Top})$

= 7569.3 N

Maximum Torsion due to wind, $T = \text{Total wind force at bottom} \cdot \text{half of mast dia at bottom}$

= 4981163.3 Nmm

Sectional Area of Mast at base, $A = \pi \cdot (D^2 - d^2) / 4$

= 11573.63 mm²

Sectional Moment of Inertia for bending at base, $I = \pi \cdot (D^4 - d^4) / 64$

= 545453482.71 mm⁴

At base, $y = D/2$

= 310.00 mm

Polar Moment of Inertia for Torsion at base, $J = \pi \cdot (D^4 - d^4) / 32$

= 1090906965.42 mm⁴

At base, $r = D/2$

= 310.00 mm

Section Modulus for bending, $z = I/y$

= 1759527.36 mm³

Direct Stress= F_y/A

= 1.48 N/mm²

Bending stress= M/z

= 134.56 N/mm²

Torsion Shear at base, $\tau_t = T \cdot r/J$

= 1.42 N/mm²

Horizontal Shear Stress due to wind, $\tau_x = k' \cdot F_x/A$

= 2.78 N/mm²

$k' = 2.0$ (Shape factor corresponding to shear stress distribution in circular/tubular section)

Stress due to Torsion Shear, $\tau_{vm} = \tau_t + \tau_x$

= 4.19 N/mm²

$0.9 \cdot f_y$

= 319.500 Mpa

Equivalent Stress, $\sigma_{e\text{ cal}} = \{\sigma_{bc\text{ cal}}^2 + 3 \cdot \tau_{vm}^2\}^{0.5}$

= 136.23 N/mm²

< $0.9 \cdot f_y$, Hence O.K.

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DESIGN FOR FOUNDATION BOLTS AND BASE PLATE

Load Data:

Design Moment at Mast Base, M	= 236754453.5 Nmm
Design Shear Force at Mast Base, F_x	= 16068.3 N
Design Dead Load (vertically downward) at Mast Base, F_y	= 17181.1 N

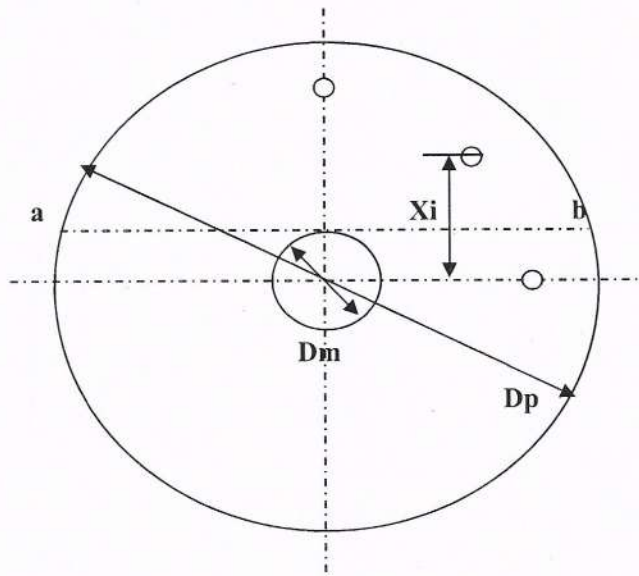
Bolt Specifications:

Size of Bolt (Coarse pitch thread)	= 39 mm dia
Property Class	= 6.8
Coarse pitch, conforming to IS: 1367 (Part-3)- 2017	
Nominal Stress Area, A_{s nom} (Ref: Table-6 of IS: 1367 (Part 3)- 2002)	= 976 mm ²
Lower yield strength, f_y (Ref: Sub Cl.5.7 Table-3 of IS: 1367 (Part 3)-2002)	= 480 N/mm ²
Min. tensile strength, R_{m,min} (Ref: Sub Cl.5.7 Table-3 of IS: 1367 (Part 3)- 2002)	= 600 N/mm ²
No. of Bolts	= 12
P.C.D. of Bolts	= 730 mm
Dia. of Bolts, D_b	= 39 mm

Base Plate Specifications:

Dia. of Base Plate, D_p	= 830 mm
Thickness of Base Plate, t	= 30 mm
Dia. of Mast at Base, D_m	= 620 mm
Yield strength of Base Plate, f_y (Grade of steel Fe-410)	= 255 N/mm ²

Design for Foundation Bolts (Close tolerance and turned bolts):



X_i = Distance of each bolts from the axis of bending

F_i = Maximum axial force (tension and compression on either side of the axis of bending) developed in bolts at different location due to bending moment (+ive or -ive)

F' = DL on each bolt (always +ive) = F_y/N = 1431.8 N

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As the number of bolts are even, at any axis of bending of the mast, the bending axis will be a line connecting two bolts and the centre of the mast base. Hence for this mast, there will be two bolts on the axis and Five bolts on either side of the axis of bending.

$$\begin{aligned} X1 &= \text{PCD}/2 \times \sin 30.0 &= 182.50 \text{ mm} \\ X2 &= \text{PCD}/2 \times \sin 60.0 &= 316.10 \text{ mm} \\ X3 &= \text{PCD}/2 \times \sin 90.0 &= 365.00 \text{ mm} \end{aligned}$$

Maximum force on bolts developed out of moment only=

$$\begin{aligned} F1 \text{ (at } X1) &= M \times X1 / 2 \times (X3^2 + 2 \times (X1^2), i=1 \text{ to } 2) &= 54053.5 \text{ N} \\ F2 \text{ (at } X2) &= M \times X2 / 2 \times (X3^2 + 2 \times (X1^2), i=1 \text{ to } 2) &= 93623.5 \text{ N} \\ F3 \text{ (at } X3) &= M \times X3 / 2 \times (X3^2 + 2 \times (X1^2), i=1 \text{ to } 2) &= 108107.1 \text{ N} \end{aligned}$$

$$\text{Maximum axial tension on bolt, } F_t \text{ max} = F_i - F' \quad = 106675.3 \text{ N}$$

$$\text{Maximum axial compression on bolt, } F_c \text{ max} = F_i + F' \quad = 109538.8 \text{ N}$$

a) Check for tension:

i) As per IS: 1367 (Part 3)- 2002

$$\begin{aligned} \text{Minimum Ultimate tensile load} &= A_s \text{ nom} \times R_m \text{ min} &= 585600.0 \text{ N} \\ (\text{Ref: Table-6 of IS: 1367 (Part 3)- 2002}) &&> 106675.3 \text{ N} \\ &&\text{Hence O.K.} \end{aligned}$$

ii) As per IS: 800- 1984

$$\begin{aligned} f_y &= 0.2\% \text{ of proof stress} &= 480 \text{ N/mm}^2 \\ f_t &= 0.7 \times \text{Tensile strength} &= 420 \text{ N/mm}^2 \end{aligned}$$

$$\begin{aligned} \sigma_{tf \text{ perm}} &= 1.0 \times f_t \times 120/235 &= 214.47 \text{ N/mm}^2 \\ (\text{Ref: Cl.11.6.2.3 of IS: 800- 2007}) && \\ \sigma_{tf \text{ cal on bolt (considering nominal stress area)}} &= F_t \text{ max} / A_s \text{ nom} &= 109.30 \text{ N/mm}^2 \\ &&< 214.47 \text{ N/mm}^2 \\ &&\text{Hence O.K.} \end{aligned}$$

As per Cl.11.6.2.3 of IS: 800- 2007

$$\text{Actual tensile stress of the bolt, } f_{tb} = T_s / A_{sb} = 106675.3 / 976 = 109.30 \text{ N/mm}^2$$

The permissible tensile stress of the bolt, $f_{atb} = 0.60 \times T_{nb} / A_{sb} =$

$$\begin{aligned} 0.60 \times 0.90 \times f_{ub} \times A_n / A_{sb} \text{ (Ref. Cl.10.3.5 of IS: 800- 2007)} & \\ \approx 0.54 \times f_{ub} = 0.54 \times 600 = &= 324.00 \text{ N/mm}^2 \\ &> 109.30 \text{ N/mm}^2 \\ &\text{Hence O.K.} \end{aligned}$$

b) Check for shear:

$$\begin{aligned} \tau_{tf \text{ perm}} &= 1.0 \times f_t \times 100/235 &= 178.72 \text{ N/mm}^2 \\ (\text{Ref: Cl.11.6.2.1 of IS: 800- 1984}) && \\ \tau_{tf \text{ cal on bolt (considering nominal stress area)}} &= (F_x / N) / A_s \text{ nom} &= 1.37 \text{ N/mm}^2 \\ &&< 178.72 \text{ N/mm}^2 \\ &&\text{Hence O.K.} \end{aligned}$$

As per Cl.11.6.2.1 of IS: 800- 2007

$$\text{Actual stress in bolt in shear, } f_{sb} = V_{sb} / A_{sb} = (16068.2688041849 / 1) = 1.37 \text{ N/mm}^2$$

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The permissible stress in bolt in shear, $f_{sb} = 0.60 \cdot V_{nsb} / A_{sb} =$

$$0.60 \cdot f_u / \sqrt{3} \cdot (\pi \cdot A_{nb} + \pi \cdot A_{sb}) = 0.35 \cdot 600 = 210.00 \text{ N/mm}^2$$

(Ref. Cl.10.3.3 of IS: 800- 2007)

$$> 1.37 \text{ N/mm}^2$$

Hence O.K.

c) Check for Combined shear & tension: (Ref: Cl.11.6.2.5 of IS: 800- 1984)

$$\sigma_{tf} \text{ cal} / \sigma_{tf} \text{ perm} + \tau_{tf} \text{ cal} / \tau_{tf} \text{ perm}$$

$$= 0.52$$

$$< 1.4$$

Hence O.K.

As per Cl.11.6.2.5 of IS: 800- 2007

$$[f_{sb} + f_{sb}]^2 + [f_{tb} + f_{tb}]^2 = [1.37 + 210]^2 + [109.3 + 324]^2 = 0.11$$

$$< 1.0$$

Hence O.K.

d) Check for bearing:

$$\sigma_{pf} \text{ perm} = 1.0 \cdot f_t \cdot 300 / 235$$

$$= 536.17 \text{ N/mm}^2$$

(Ref: Cl.11.6.2.2 of IS: 800- 2007)

$$\sigma_{tf} \text{ cal on bolt (considering nominal stress area)} = F_c \text{ max} / A_s \text{ nom}$$

$$= 112.23 \text{ N/mm}^2$$

$$< 536.17 \text{ N/mm}^2$$

Hence O.K.

As per Cl.11.6.2.2 of IS: 800- 2007

Actual stress of bolt in bearing on any plate, $f_{pb} = V_{sb} / A_{pb}$

$$= 109538.8 / 976 = 112.23 \text{ N/mm}^2$$

The permissible bearing stress of the bolt/ plate,

$$= 0.6 \cdot 2.5 \cdot k_b \cdot d \cdot t \cdot f_u / A_{pb}$$

(Ref. Cl.10.3.4 of IS: 800- 2007)

k_b is smaller of $e/3d_o$, $p/3d_o - 0.25$, f_{ub}/f_u , 1.0

where,

$$e = 50 \text{ mm}$$

$$p = 191.11 \text{ mm}$$

$$d_o = 42 \text{ mm}$$

Therefore, $e/3d_o =$

$$50 / (3 \cdot 42) = 0.40$$

$$p/3d_o - 0.25 =$$

$$191.11 / (3 \cdot 42) - 0.25 = 1.27$$

$$f_{ub}/f_u =$$

$$480 / 600 = 0.80$$

$$= 1.0$$

$$\text{Hence, } k_b = 0.40$$

$$\text{Now, } 0.6 \cdot 2.5 \cdot k_b \cdot d \cdot t \cdot f_u / A_{pb} =$$

$$0.60 \cdot 2.5 \cdot 0.4 \cdot 39 \cdot 30 / 976 = 431.56 \text{ N/mm}^2$$

$$> 112.23 \text{ N/mm}^2$$

Hence O.K.

e) Check for embedment length:

(Ref: Cl.26.2.1.1 of IS: 456- 2000)

Considering Grade of Concrete

$$= \text{M20}$$

Design bond stress, τ_{bd} in tension

$$= 1.2 \text{ N/mm}^2$$

Design bond stress, τ_{bd} in compression

$$= 1.50 \text{ N/mm}^2$$

(Though the bolt surface will have rough surface, no increment allowance on the τ_{bd} value is taken in consideration.)

i) In Tension:

$$\text{Straight length required for each bolt, } L = 1.5 \cdot F_t \text{ max} / \pi \cdot D_b \cdot \tau_{bd}$$

$$= 1088 \text{ mm}$$

i) In Compression:

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Straight length required for each bolt, $L = 1.5 \cdot F_c \max / \pi \cdot D_b \cdot \tau_{bd}$ = 894 mm
Using anchor plate at bottom of bolt, straight length required may be reduced considering the
8x bolt dia (Ref: Cl.26.2.2.1 of IS: 456- 2000) = 312 mm
Hence bolt length required will be = 776 mm
Provide foundation bolt with embedment length of = 1000 mm
Hence O.K.

Design for Base Plate:

a) Check for Max. Shear stress:

Max. Bending Moment at Mast Base, M = 236754453.5 Nmm
Max. Permissible Shear stress = $1.0 \cdot 0.4 f_y$ = 102.00 N/mm²
(Ref: Cl.11.4.2 of IS: 800- 2007)
Max. Calculated shear on the base plate at the base of the mast, $\tau_{\max} = 4 \cdot M / (\pi \cdot D_p^2 \cdot t)$
= 14.59 N/mm²
< 102.00 N/mm²
Hence O.K.

b) Check for Max. Bending stress:

Approximate Max. Bending Moment at the junction of the Base Plate and Mast (on line a-b)
 $M_{\max'} = F_3 \cdot (X_3 - D_m/2) + 2 \cdot F_2 \cdot (X_2 - D_m/2)$ = 7087958.0 Nmm
The breadth of the section of the base plate at its junction with the base plate,
 $B = 2 \cdot [(D_p/2)^2 - (D_m/2)^2]^{0.5}$ = 551.82 mm
If required thickness of the base plate is t_1 , the max moment can be taken by the plate,
 $M_{\max 1} = 1.0 \cdot 0.66 \cdot f_y \cdot B/6 \cdot t_1^2$
Now equating $M_{\max'}$ and $M_{\max 1}$, $t_1 = \{M_{\max'} / (1.0 \cdot 0.66 \cdot f_y \cdot B/6)\}^{0.5}$
= 21.40 mm
< 30.00 mm
Hence O.K.

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PART - B

STRUCTURAL DESIGN CALCULATIONS FOR RCC FOUNDATION

DESIGN BASIS :

The foundation have been designed considering "Limit State Method" on the basis of soil data supplied by the client and other loading data obtained from the analysis results. The superstructure has been analysed with the dead load and lateral load as per IS: 875 (Part 3)-2015, to derive the loading pattern and their values on the foundation.

DESIGN PARAMETERS :

1. Load Bearing Capacity of soil	= 4.0 T/m ²
Soil Type : Black Cotton Soil (Data Supplied by the client in absence of Soil Test Report)	
2. Grade of Concrete	= M20
3. Grade of Steel	= Fe500
4. Clear cover to main reinforcement	= 50 mm
5. Permissible factor of safety against overturning and sliding	= 1.5
6. Difference between G.L. and top of pedestal	= 0.3 m
7. Depth of foundation below GL.	= 1.70 m
8. ϕ value for soil (assumed in absence Soil Test Report)	= 15 degree
9. Unit wt. of concrete	= 2.5 T/m ³
10. Unit wt. of soil (assumed in absence Soil Test Report)	= 1.8 T/m ³

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DESIGN CALCULATION FOR RCC FOUNDATION

Type of Foundation is Shallow Foundation with square base

Net Bearing Capacity of soil, p_{net}	= 4.0 T/m ²
Gross Bearing Capacity of soil, $p_{gross} = p_{net} + \gamma \cdot h$	= 7.06 T/m ²
As the value of the governing wind force is more than 25% of dead load, Permissible Bearing Capacity of soil will be 25% more, Permissible bearing capacity of soil due to wind may be considered as, p_{perm} (Ref: Cl.15.1.7 of IS: 1904- 1986)	= 8.06 T/m ²
Total Horizontal shear forces at base of mast (WL)= F_x	= 16068.3 N
Total Vertical forces at base of mast (DL)= F_y	= 17181.1 N
Moment on vertical plane, at base of mast (WL)= M_z	= 236754453.5 Nmm
Characteristic strength of Concrete, f_{ck}	= 20 N/mm ²
Characteristic strength of Steel, f_y	= 500 N/mm ²

Depth of foundation below GL., h	= 1.70 m
Difference between GL. and top of RCC pedestal, h_1	= 0.3 m
Size of Footing, width, B	= 3.60 m
length, L	= 3.60 m
Depth of footing slab, D	= 0.35 m
Size of pedestal, width, B_p	= 1.20 m
length, L_p	= 1.20 m

Dead Load at Foundation bottom:

a) Self weight of pedestal= $B_p \cdot L_p \cdot (h + h_1 - D) \cdot 2.5$	= 5.9 T
b) Self weight of footing= $B \cdot L \cdot D \cdot 2.5$	= 11.34 T
Total weight of RCC foundation, W_{rcc}	= 17.28 T
c) Weight of Backfill on footing, $W_{soil} = (B \cdot L - B_p \cdot L_p) \cdot (h - D) \cdot 1.8$	= 27.99 T
Total weight (submerged) at the centre of base, $W_{sub} = F_y + (W_{rcc}/2.5) \cdot (2.5 - 1) + (W_{soil}/1.8) \cdot (1.8 - 1)$	= 24.53 T
Total Gross weight at the centre of base, $W_{gr} = F_y + W_{rcc} + W_{soil}$	= 46.99 T

CHECK FOR SLIDING:

(Ref: Cl.20.1 of IS: 456- 2000)

Sliding Force, F_x	= 1.61 T
Resistance force against sliding, $F_r = W_{sub} \cdot \tan \phi$	= 6.57 T
F.S. against Sliding, F_r/F_x	= 4.09
	> 1.5
	Hence O.K.

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CHECK FOR OVERTURNING:

(Ref: Cl.20.1 of IS: 456- 2000)

Overturning Moment (about footing edge), $M_o = M_z + F_x \cdot (h + h_1)$ = 26.89 T-m

Moment of resistance against overturning, $M_r = W_{sub} \cdot B/2$ = 44.15 T-m

F.S. against overturning, M_r/M_o = 1.64

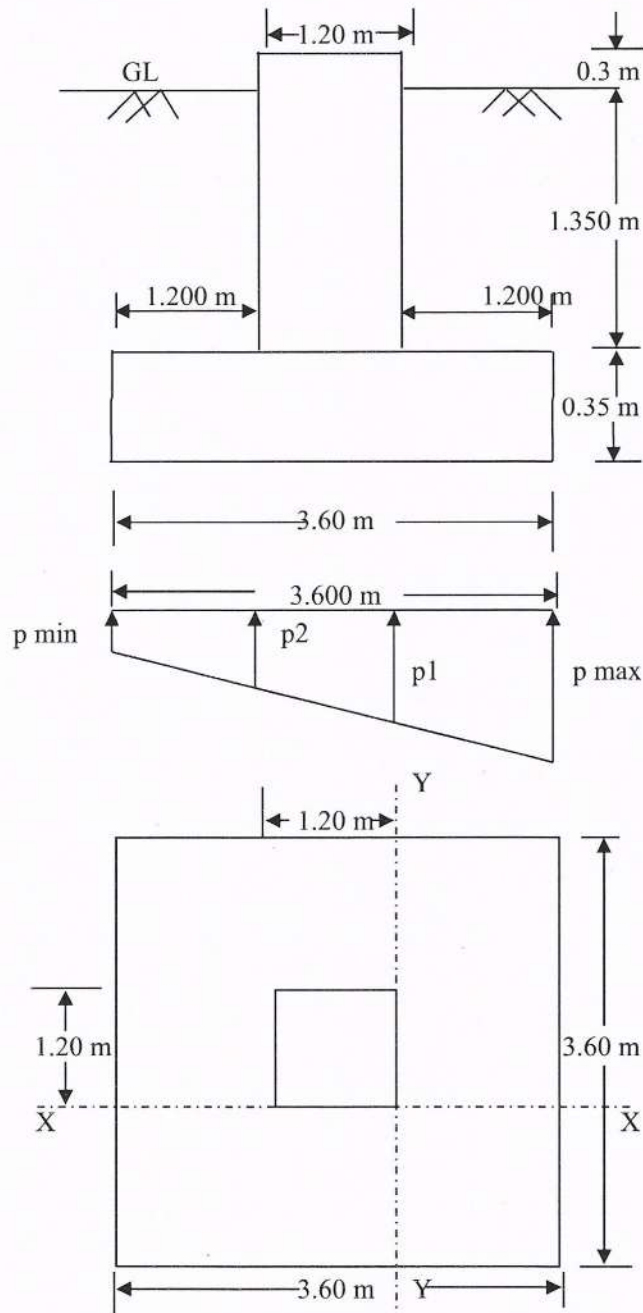
> 1.5

Hence O.K.

CALCULATION FOR MAXIMUM SOIL PRESSURE:

Area of footing, $A = L \cdot B$

= 13.0 m²



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Total Overturning Moment, M_o = 26.89 T-m
Total Vertical Load, V = 46.99 T

About Y-Y axis:

Eccentricity of resultant vertical forces = 0.572
< $B/6$

Therefore Base pressure distribution is Trapezoidal

Width of footing in contact with soil = 3.600 m
At one edge Gross Max. Pressure, p_{max} = 7.08 T/m²
> 8.06 T/m²
Hence O.K.
At one edge Gross Min. Pressure, p_{min} = 0.17 T/m²
Pressure at pedestal face, p_1 = 4.78 T/m²
Pressure at pedestal face, p_2 = 2.47 T/m²
Factored moment on footing at the face of pedestal, M_u = 6.82 T-m per m
 $M_u = 0.138 \cdot f_{ck} \cdot b \cdot d^2$
Effective depth required, $d_{reqd.}$ = 157.19 mm
< 294 mm
Hence O.K.

About X-X axis:

Eccentricity of resultant vertical forces = 0.572
< $B/6$

Therefore Base pressure distribution is Trapezoidal

Width of footing in contact with soil = 3.600 m
At one edge Gross Max. Pressure, p_{max} = 7.08 T/m²
< 8.06 T/m²
Hence O.K.
At one edge Gross Min. Pressure, p_{min} = 0.17 T/m²
Pressure at pedestal face, p_1 = 4.78 T/m²
Pressure at pedestal face, p_2 = 2.47 T/m²
Factored moment on footing at the face of pedestal, M_u = 6.82 T-m per m
 $M_u = 0.138 \cdot f_{ck} \cdot b \cdot d^2$
Effective depth required, $d_{reqd.}$ = 157.19 mm
< 282 mm
Hence O.K.

Design of bottom bar in base:

About Y-Y axis:

Factored moment on footing at the face of pedestal, M_u = 6.82 T-m per m
(Tension at bottom)

$M_u / b d^2$ = 0.789

From Table 2 of SP- 16, p_t reqd. = 0.190%

Provide 12 mm dia bars @ 200 mm c/c along x-x axis at bottom
 p_t provided = 0.203%

> 0.190%
Hence O.K.

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About X-X axis:

Factored moment on footing at the face of pedestal, M_u	= 6.82 T-m per m
(Tension at bottom),	
M_u/bd^2	= 0.858
From Table 2 of SP- 16, pt reqd.	= 0.210%
Provide 12 mm dia bars @ 200 mm c/c along y-y axis at bottom	
pt provided	= 0.212%
	> 0.210%
	Hence O.K.

Design of top bar in base:

About Y-Y axis:

At one edge Gross Min. Pressure, p min	= 0.1679 T/m ²
Factored moment on footing at the face of pedestal, M_u	= 0.00 T-m per m
(Tension at top)	
M_u/bd^2	= 0.000
From Table 2 of SP- 16, pt reqd.	= 0.070%
Provide 10 mm dia bars @ 200 mm c/c along x-x axis at top	
pt provided	= 0.141%
	> 0.070%
	Hence O.K.

About X-X axis:

At one edge Gross Min. Pressure, p min	= 0.1679 T/m ²
Factored moment on footing at the face of pedestal, M_u	= 0.00 T-m per m
(Tension at top)	
M_u/bd^2	= 0.000
From Table 2 of SP- 16, pt reqd.	= 0.070%
Provide 10 mm dia bars @ 200 mm c/c along y-y axis at top	
pt provided	= 0.147%
	> 0.070%
	Hence O.K.

CHECK FOR ONE WAY SHEAR:

(Ref: Cl.40.4 of IS: 456- 2000)

Critical section is at a distance 'd' from the face of pedestal

About Y-Y axis:

Factored Shear force, V_{u1}	= 8.44 T per m
Effective depth at critical section, d	= 294 mm
Nominal Shear stress, $\tau_{v1} = V_{u1}/bd$	= 0.287 N/mm ²
Percent of tensile steel, pt provided	= 0.203%
Design Shear Strength of M20 concrete = τ_c	= 0.320 N/mm ²
(Ref: Table 19 of IS: 456- 2000)	> 0.287 N/mm ²
	Hence O.K.
Maximum Shear Strength, $\tau_c \text{ max}$ (Ref: Table 20 of IS: 456- 2000)	= 2.8 N/mm ²
Design Shear Strength of M20 concrete = τ_c	= 0.320 N/mm ²
	< 2.8 N/mm ²
	Hence O.K.

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About X-X axis:

Factored Shear force, V_{u1}		= 8.54 T per m
Effective depth at critical section, d		= 282 mm
Nominal Shear stress, $\tau_{v1} = V_{u1}/bd$		= 0.303 N/mm ²
Percent of tensile steel, p_t provided		= 0.212%
Design Shear Strength of M20 concrete	= τ_c	= 0.325 N/mm ²
(Ref: Table 19 of IS: 456- 2000)		> 0.303 N/mm ²
		Hence O.K.
Maximum Shear Strength, $\tau_{c \text{ max}}$ (Ref: Table 20 of IS: 456- 2000)		= 2.8 N/mm ²
Design Shear Strength of M concrete	= τ_c	= 0.325 N/mm ²
		< 2.8 N/mm ²
		Hence O.K.

CHECK FOR PUNCHING SHEAR:

(Ref: Cl.31.6.3 of IS: 456- 2000)

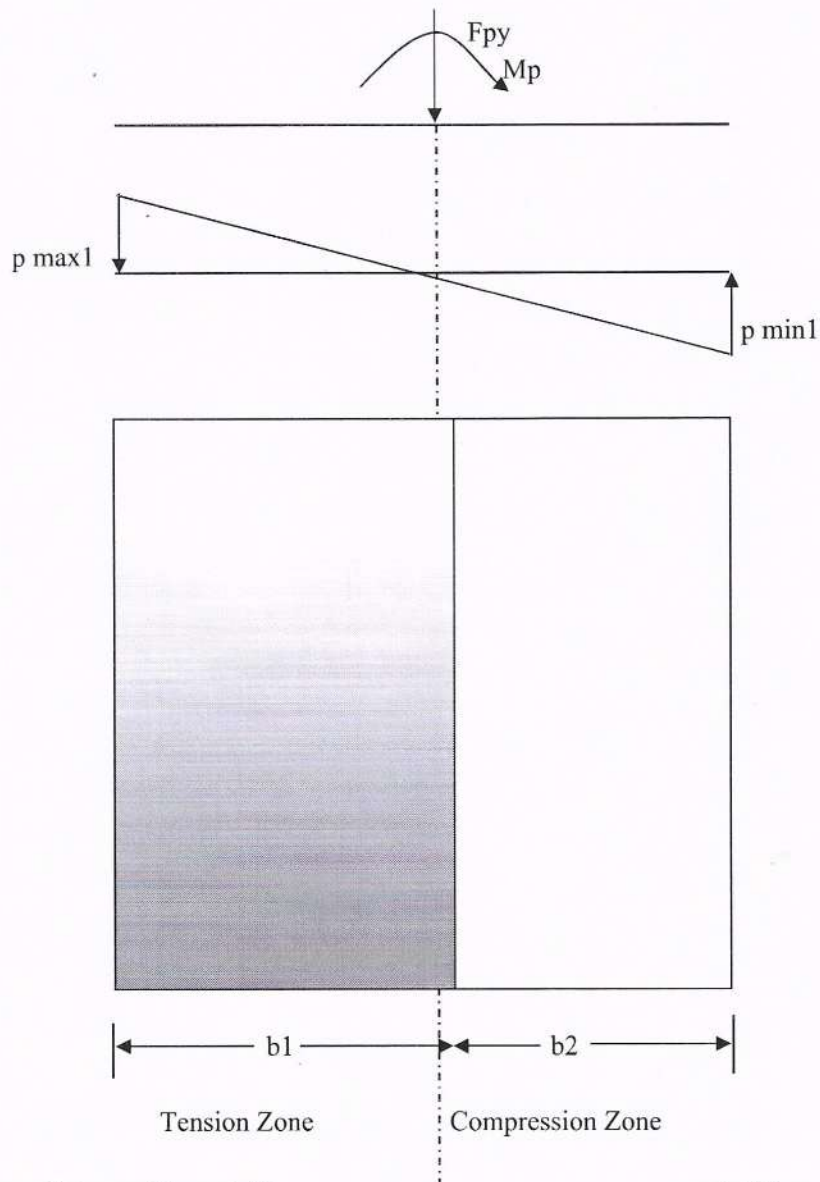
Critical section is at a distance ' $d/2$ ' from the periphery of pedestal

Factored Shear force, V_{u2}		= 97.72 T
Nominal Shear stress, $\tau_{v2} = V_{u2}/b_{od}$		= 0.556 N/mm ²
$k_s = (0.5 + \beta_c)$		= 1.00
β_c = Length of shorter side of pedestal/Length of longer side of pedestal		= 1.00
$\tau_c = 0.25 \cdot \sqrt{f_{ck}}$		= 1.12
Design Shear Strength of M20 concrete	= $\tau_c' = k_s \cdot \tau_c$	= 1.12 N/mm ²
		> 0.556 N/mm ²
		Hence O.K.

DESIGN FOR PEDESTAL:

Sectional Moment of Inertia of pedestal= $I_p = L_p \cdot B_p^3/12$		= 0.173 m ⁴
$y_{\text{max}} = L_p/2$		= 0.600 m
Section Modulus of pedestal= $z_p = I_p/y_{\text{max}}$		= 0.288 m ³
Cross Sectional Area of pedestal= $A_p = L_p \cdot B_p$		= 1.44 m ²
Axial force, $F_{py} = F_y + s/w$ of pedestal		= 6.758 T
Moment at pedestal bottom, $M_p = M_z + F_x \cdot (h + h_1 - D)$		= 26.327 T-m
Maximum stress in pedestal, $p_{\text{max}1} = F_{py}/A_p + M_p/z_p$		= 96.105 T/m ² (Comp)
Minimum stress in pedestal, $p_{\text{min}1} = F_{py}/A_p - M_p/z_p$		= 86.719 T/m ² (Ten)

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$$b1 = Lp * p_{max1} / (p_{max1} + p_{min1})$$

$$= 0.631 \text{ m}$$

$$b2 = Lp - b1$$

$$= 0.569 \text{ m}$$

Design for pedestal Reinforcement:

Check for tension:

$$\text{Sectional Area of pedestal in tension zone} = A_t = b1 / Lp * A_p$$

$$= 0.757 \text{ m}^2$$

$$\text{Total tension force, } T = 0.5 * A_t * p_{max1} =$$

$$= 36.374 \text{ T}$$

$$= 363741.85 \text{ N}$$

Considering Tensile Strength of steel, σ_t

$$= 435 \text{ N/mm}^2$$

Area of steel required, $A_{st} = T / \sigma_t$

$$= 836.19 \text{ mm}^2$$

Provide 20 nos.

12 mm dia bars

Total Area provided in tension zone, A_{st} provided

$$= 1130.97 \text{ mm}^2$$

$$> 836.19 \text{ mm}^2$$

Hence O.K.

Total Area of reinforcement provided, A_{st} provided

$$= 2261.95 \text{ mm}^2$$

Minimum Area of reinforcement required = 15% of area of pedestal

$$= 2160.00 \text{ mm}^2$$

(Ref: Cl.26.5.3.1 (h) of IS: 456- 2000)

Hence O.K.

**BPP
HIGH MAST**

Check for compression:

$$\begin{aligned}\text{Sectional Area of pedestal in compression zone} &= A_c = b^2/L_p * A_p &= 0.683 \text{ m}^2 \\ \text{Total compression force, } C &= 0.5 * A_c * p_{\min} &= 29.616 \text{ T} \\ & &= 296160.72 \text{ N}\end{aligned}$$

Compressive force is to be taken by the concrete itself alone in the compression zone.

$$\begin{aligned}\text{Compressive force that can be taken by the concrete itself alone in the compression zone} &= 0.4 * f_{ck} * A_c &= 5464279.55 \text{ N} \\ & &> 296160.72 \text{ N} \\ & &\text{Hence O.K.}\end{aligned}$$

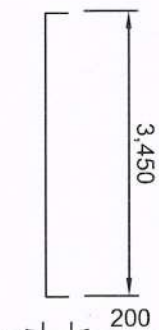
Provide 8 mm dia bars @ 190 mm c/c as tie for pedestal reinforcement

$$\begin{aligned}\text{(Ref: Cl.26.5.3.2 (c) of IS: 456- 2000)} & &< 1200 \text{ mm} \\ & &< 192 \text{ mm} \\ & &< 300 \text{ mm} \\ & &\text{Hence O.K.}\end{aligned}$$

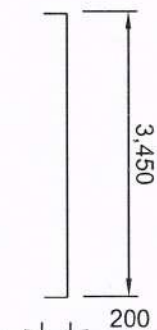
BILL OF MATERIALS.

1. EXCAVATION = 35.20 CU.M.
2. CONC. VOL. = 6.91 CU.M.
3. PCC VOL. = 0.13 CU.M.
4. BRICK REED. FOR SOLING = 500 NOS.
5. HALF INCH SIZE STONE CHIPS = 4.03 CU.M.
6. COARSE SAND = 2.02 CU.M.
7. CEMENT = 1.00 CU.M. = 29 BAGS
8. STEEL Fe 500 = 316.62 KG.
9. BAR BENDING SCHEDULE :-

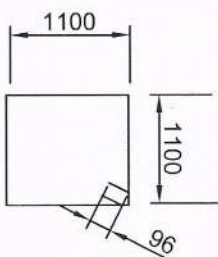
A. 12 mm. DIA. TOR BAR, NOS. = 38 = 129.89 KGS.



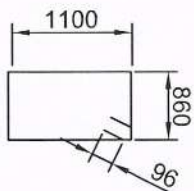
B. 10 mm. DIA. TOR BAR, NOS. = 38 = 90.20 KGS.



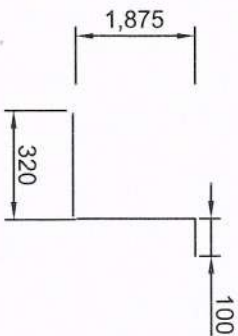
D. 8 mm. DIA. RING, NOS. = 11 = 20.52 KGS.



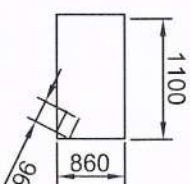
E. 8 mm. DIA. RING, NOS. = 11 = 17.63 KGS.



C. 12 mm. DIA. TOR ROD, NO. OF = 20 = 40.75 KGS.



F. 8 mm. DIA. RING, NOS. = 11 = 17.63 KGS.



NOTE :-

1. ALL DIMENTIONS ARE IN MM UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE M20
3. REINFORCEMENT SHALL BE OF HIGH STRENGTH DEFORMED BARS (TMT) BARS OF GRADE Fe 500 CONFORMING IS: 1786
4. MINIMUM CLEAR COVER FOR MAIN REINFORCEMENT SHALL BE 50MM
5. MINIMUM CLEAR COVER FOR PEDESTAL REINFORCEMENT SHALL BE 50 MM
6. USE STEEL CHAIRS TO KEEP TOP REINFORCE - MENT IN POSITION.

Checked & Vetted

Dr. Partha Ghosh
B.E. (Civil), M.E. (Structural Engg.), Ph.D. (Engg.)
Professor
Construction Engg. Department
Jadavpur University, Kolkata-700 106

WIND SPEED = 180 Km/Hr (50 m/s).

B. P. PROJECTS PVT. LTD.

167 A, VIVEKANANDA ROAD

KOLKATA - 700 006.

PROJECT :-

TITLE :- FOUNDATION DETAILS FOR 22 MTR. SIGNAGE MAST (INLAND)

(HAVING 4 $\frac{1}{m^2}$ SOIL BEARING CAPACITY)

DESIGN	MB	24.07.25	THIS DRAWING IS THE PROPERTY OF B.P. PROJECTS PVT. LTD., KOLKATA AND MUST NOT BE COPIED BY ANY PERSON WITHOUT WRITTEN CONSENT OF THE AUTHORITY.
DRN.	MB	24.07.25	
CHKD	BPP	24.07.25	
APPD	BPP	24.07.25	



DRG. NO. :- Rev. No. 0

MARKED. REVISIONS.

DATE.

APPD.

BPP

24.07.25

SCALE

N. T. S.

CLIENT: BHARAT PETROLEUM CORPORATION LIMITED